

TRANSASIA BIO-MEDICALS LTD

EM 200 (Clinical Chemistry Analyzer)

ASTM - Host Interface Document

Version: 2.0

VERSION CONTROL:

Publication Reference No.:	Version 2.0
Date:	Nov 2008
Pages Affected:	All

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This manual was created by the Transasia Bio-Medicals Ltd Research and Development Department.

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OVERVIEW:

Purpose:

This Document details the specifications for ASTM Host Interconnection and communication for Clinical Chemistry analyzer by Transasia Bio-Medicals Lid.

Scope:

Detailed information on operation of the system is beyond the scope of this document. The information offered here is strictly to aid programmers in grasping very basic operational features of the analyzer.

Conventions:

This document consists primarily of a series of examples that show the information needed to successfully interface to the system. The basic concept of data transfer in this interface is the exchange of data and control frames between the host system and the analyzer.

Primary Reference:

E1381 – 02

Low-Level Protocol to Transfer Messages between Clinical Laboratory Instruments and Computer Systems.

E1394 – 97

Transferring Information Between Clinical Instruments and Computer Systems.

INTERCONNECTION DIAGRAM:

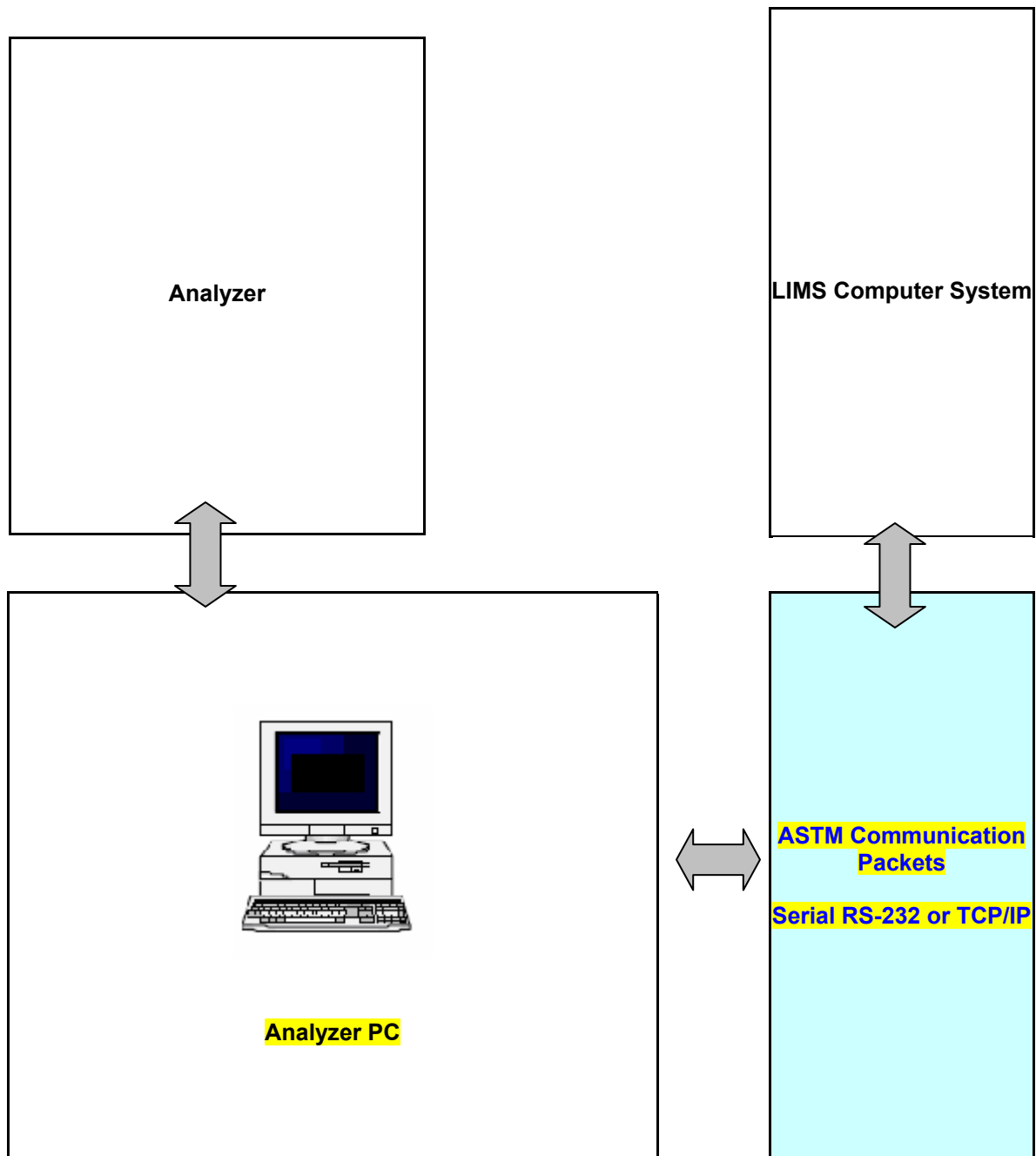


Fig (2.0) ASTM Interconnection with LIMS Computer System

OSI (Open System Interconnection) Layers:

Application Layer	
Presentation Layer	Application Software
Session Layer	
Transport Layer	MsComm (VBA Component) or TCP/IP (Win socket)
Network Layer	
DataLink Layer	ANSI X3.4-1986 standards. / IEEE 802.3 standards. <u>Frame Structure:</u> <STX> FN text <ETX> C1 C2 <CR><LF>: End Frame Or <STX> FN text <ETB> C1 C2 <CR><LF>: Intermediate frame. (For messages greater than 1024 characters) FN = Frame No.
Physical Layer	<u>RS232 / Fast Ethernet (TCP/IP)</u> Electrical Characteristics: EIA-232-D-1986 standards. Serial: 9600 BAUD, 8-bit, No Parity, 1 Stop Bit. TCP/IP: 10MBPs Connectors: DB9 / DB25 or RJ45.

PHYSICAL LAYER:

1. ASTM provides an option of RS232C : Serial Communication for the user to connect with LIMS Computer System.

RS232C (Serial Communication)

Communication Specifications:

Transmission Method	RS232C asynchronous, half duplex.
Transmission Rate	1200, 2400, 4800, <u>9600</u> , 19200 bps
Transmission Code	ASCII
Data Length	7 bits, <u>8 bits</u>
Parity	Even, Odd, <u>None</u>
Stop Bit	1 bit , 2 bit

Connections:



Connector:

D-SUB 9-Pin Connector (Male) on Standard PC

D-SUB 9-Pin Connector (Female): for Communication Cable.

Pin Assignment: Communication Cable

D-SUB 9-Pin			D-SUB 9-Pin	
Frame GND	1		1	Frame GND
Receive Data	2	⇐	3	Transmit Data
Transmit Data	3	⇒	2	Receive Data
Data Terminal Ready	4	⇐	6	Data Set Ready
Signal GND	5	↔	5	Signal GND
Data Set Ready	6	⇒	4	Data Terminal Ready
Request to Send	7	⇒	8	Data Set Ready
Data Set Ready	8	⇐	7	Request to Send
NC	9		9	NC

NOTE: For 25-Pin Connector consider following Pin details:

Pin		Description
1		Frame GND
2	TXD	Transmit Data
3	RXD	Receive Data
4	RTS	Request to Send
5	CTS	Clear to Send
6	DSR	Data Set Ready
7	GND	Signal GND
20	DTR	Data Terminal Ready

MODIFYING HOST SETTINGS:

Go to Settings Window and select “Host Connection”

System Parameters Carryover Pairs Test Sequence Serum Indices Rerun Flags User Rights

Left Sidebar:

- Patient Entry(F2)
- Test Parameters(F3)
- Profiles / Calc(F4)
- QC/Calibrations(F5)
- Consumables(F6)
- Status Monitor(F7)
- Search(F8)
- Reports(F9)
- Master
- Utility(F11)
- Service Check
- Maintenance(F12)
- Settings
- Shut Down

Main Content Area:

Laboratory Name : TRANSASIA BIO-MEDICALS LTD*

Default Language : English Clear Screen Upon Save : Yes

Printing Mode : OFF Patient Report : ☒

Print Negative Result : ☐ Confirmation Message : ☐

Analysar Port : COM1 **Host Connection : ☒**

Open Channel Test : Open

RCT Temperature : 37.0 RCT Temperature Range : 0.2

RGT Temperature : 8.0 RGT Temperature Range : 4.0

Sample Barcode : ☒ Reagent Barcode : ☒

ISE Module : ☒

Minimum Cell Blank : 0.03 Maximum Cell Blank : 0.10

Indication : Set Host Connection status ON/OFF

Buttons: [Icon] [Icon] [Icon] [Icon] [SAVE] [CLEAR] [EDIT] [DELETE]

Go to Host Setting. If 'Host Setting' menu is not visible click on Settings again .
There are two modes of Communication

1. RS232 for Serial Communication

Select RS232 Option for Serial Communication. Select 'Send Control Result To Host'

If want to send Control Result. Select 'Send To Host' option for sending result with selected flag.

System Parameters **Carryover Pairs** **Test Sequence** **Rerun Flags** **User Rights** **Host Setting**

RS 232C

Port No. : COM8

Baud Rate : 9600

Parity : NONE

Data Length : 8

Stop Bits : 1

TCPIP

IP Address : 127.0.0.1

Port No. : 12377

CONNECT **DISCONNECT**

☒ Send Control Result To Host

PROTOCOL PARAMETERS

Delimiter	Actual Value	Recommended Value
STX	2	2
ETX	3	3
ACK	6	6
NAK	21	21
ENQ	5	5
EOT	4	4
Field Delm	124	124
Repeat Delm	96	96
Component Delm	94	94
Escape Delm	38	38

Set Flags :

Send To Host	Flags Description	Host Flags
☒	#	A
☐	-	A
☐	F	A
☐	+1SD	A
☐	A* (Cal A Absent)	A
☒	NOCAL	A
☐	?SD	A
☐	V-D	A
☐	V-I	A
☐	MONO	A
☐	PD	A

Indication : Select communication mode and then save changes.

CHECK CONNECTION **SAVE** **CLEAR** **EDIT**

2. TCP/IP

Select TCP/IP Option for TCP/IP Communication. Enter the IP address of the machine in which LIMS is running. After clicking on 'CONNECT' LIMS Connected successfully message will come

The screenshot displays the 'Host Setting' window in the LIMS software. The interface includes a sidebar with various system functions and a main area for configuring communication parameters.

System Parameters:

- RS 232C:** Port No. (COM8), Baud Rate (9600), Parity (NONE), Data Length (8), Stop Bits (1).
- TCP/IP:** IP Address (192.168.8.105), Port No. (123). Buttons: CONNECT, DISCONNECT.
- ☒ Send Control Result To Host

PROTOCOL PARAMETERS:

Delimiter	Actual Value	Recommended Value
STX	2	2
ETX	3	3
ACK	6	6
NAK	21	21
ENQ	5	5
EOT	4	4
Field Delm	124	124
Repeat Delm	96	96
Component Delm	94	94

Flags Description:

Flags Description	Host Flags
F	A
+1 SD	A
A* (Cal A Absent)	A
☒ NOCAL	A
?SD	A
V-D	A
V-I	A
MONO	A
PD	A

Indication: Select communication mode and then save changes.

Buttons: CHECK CONNECTION, SAVE, CLEAR, EDIT.

MultiXL Dialog: LIMS is Connected Successfully at 192.168.8.105. OK

If LIMS is not connected Successfully then click on 'Check Connection' and check the status of Communication.

System Parameters Carryover Pairs Test Sequence Rerun Flags User Rights **Host Setting**

Patient Entry(F2)
Test Parameter(F3)
Profiles / Calc(F4)
QC/Calibration(F5)
Consumables(F6)
Status Monitor(F7)
Search(F8)
Reports(F9)
Master
Utility(F11)
Service Check
Maintenance(F12)
Settings
Shut Down

RS 232C

Port No. : COM8
Baud Rate : 9600
Parity : NONE
Data Length : 8
Stop Bits : 1

TCPIP

IP Address : 192.168.8.105
Port No. : 123
CONNECT **DISCONNECT**

☒ Send Control Result To Host

Indication : Communication Ok

CHECK CONNECTION **SAVE** **CLEAR** **EDIT**

PROTOCOL PARAMETERS

Delimiter	Actual Value	Recommended Value
STX	2	2
ETX	3	3
ACK	6	6
NAK	21	21
ENQ	5	5
EOT	4	4
Field Delm	124	124
Repeat Delm	96	96
Component Delm	94	94
	38	38

MultiXL

Communication Ok
OK

Flags	Description	Host Flags
#		A
-		A
F		A
+1 SD		A
A* (Cal A Absent)		A
☒	NOCAL	A
?SD		A
V-D		A
V-I		A
MONO		A
PD		A

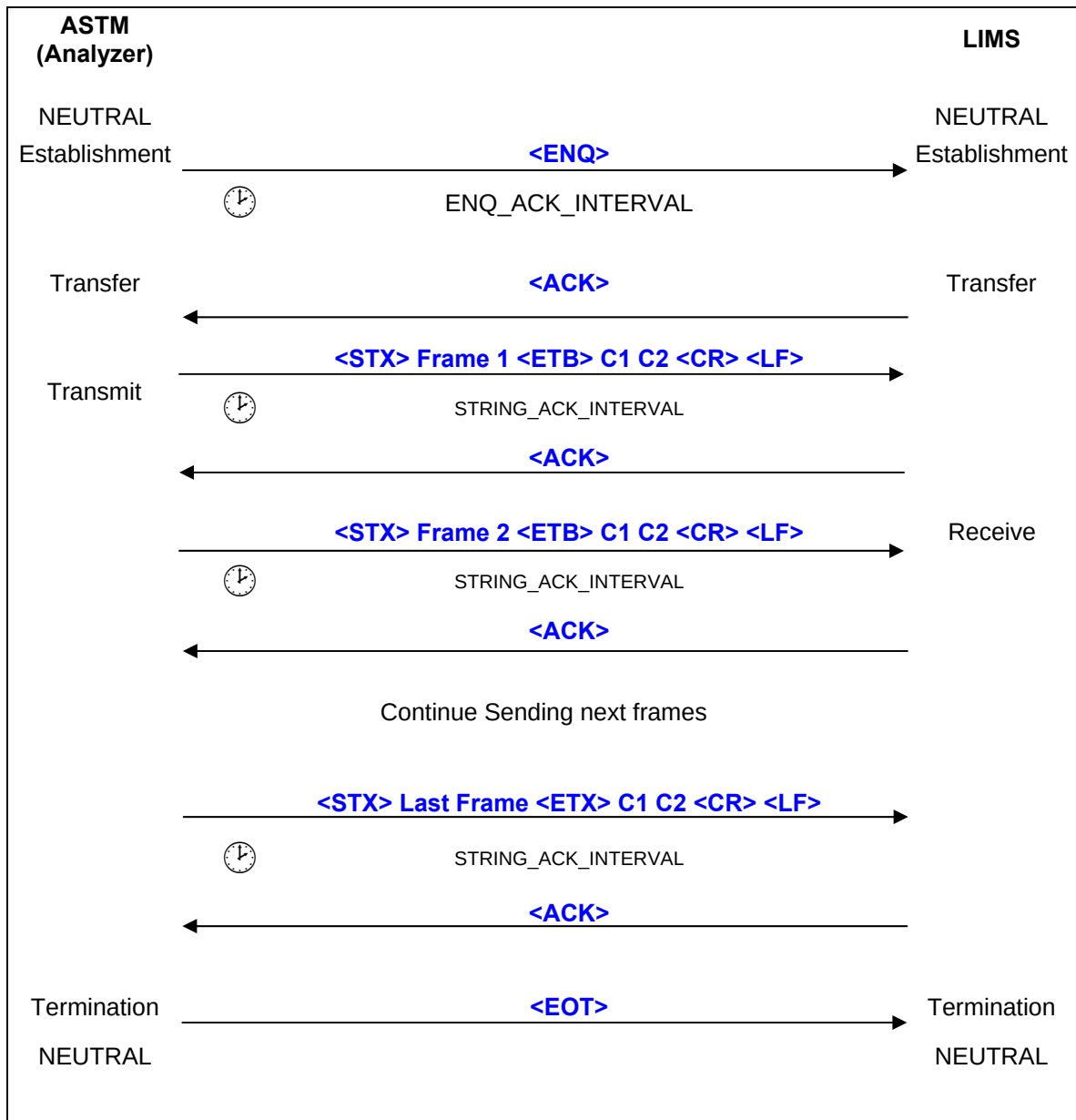
ASTM Lower Level Communication Methodology:

Item	Method	Explanation
Frame Configurations	For Middle Frame <STX> FN text <ETB> C1 C2 <CR><LF> For Last Frame <STX> FN text <ETX> C1 C2 <CR><LF>	Control character (characters enclosed in <>): <STX> is control character (HEX 02) <ETB> is control character (HEX 17) <CR> is control character (HEX 0D) <LF> is control character (HEX 0A) <ETX> is control character (HEX 03) FN: FN is a single ASCII number. FN indicates the sequence number for a frame (the frame number modulus 8). Frames of a single transmission phase are consecutively numbered beginning with 1, so FN runs from 1 to 7, then continues with 0, 1, and so on. Text: the data content of a frame (maximum 1024 characters). Records are sub-divided into intermediate (middle) frames with 1024 or fewer characters. Text is part of a split message. C1 and C2: When 1 byte resulting from adding each byte, FN to <ETB> for the middle frame and FN to <ETX> for the last frame, is expressed in hexadecimal, the upper character (161) is C1 and the lower character (160) is C2. Characters used are '0' to '9' or 'A' to 'F'
Frame Character Configuration of Text	Characters other than <SOH><STX><ETX> <EOT><ENQ><ACK> <DLE><NAK><SYN> <ETB><LF><DC1> <DC2><DC3><DC4>	<SOH> is control character (HEX 01) <EOT> is control character (HEX 04) <ENQ> is control character (HEX 05) <ACK> is control character (HEX 06) <DLE> is control character (HEX 10) <NAK> is control character (HEX 15) <SYN> is control character (HEX 16) <DC1> ~ <DC4> is control character (HEX 11 ~ 14)
Maximum Length of the Frame	1024 characters	For one frame, maximum of 1024 characters for text, 7 characters for frame control characters. Messages equal to or less than 1024 characters are transmitted as one final frame. Messages greater than 1024 characters are split into frames that have character lengths that fall within the 1024-character limit. The only or final remaining frame becomes the last frame and is indicated by <ETX>. All others are intermediate (middle) frames and are indicated by <ETB>.

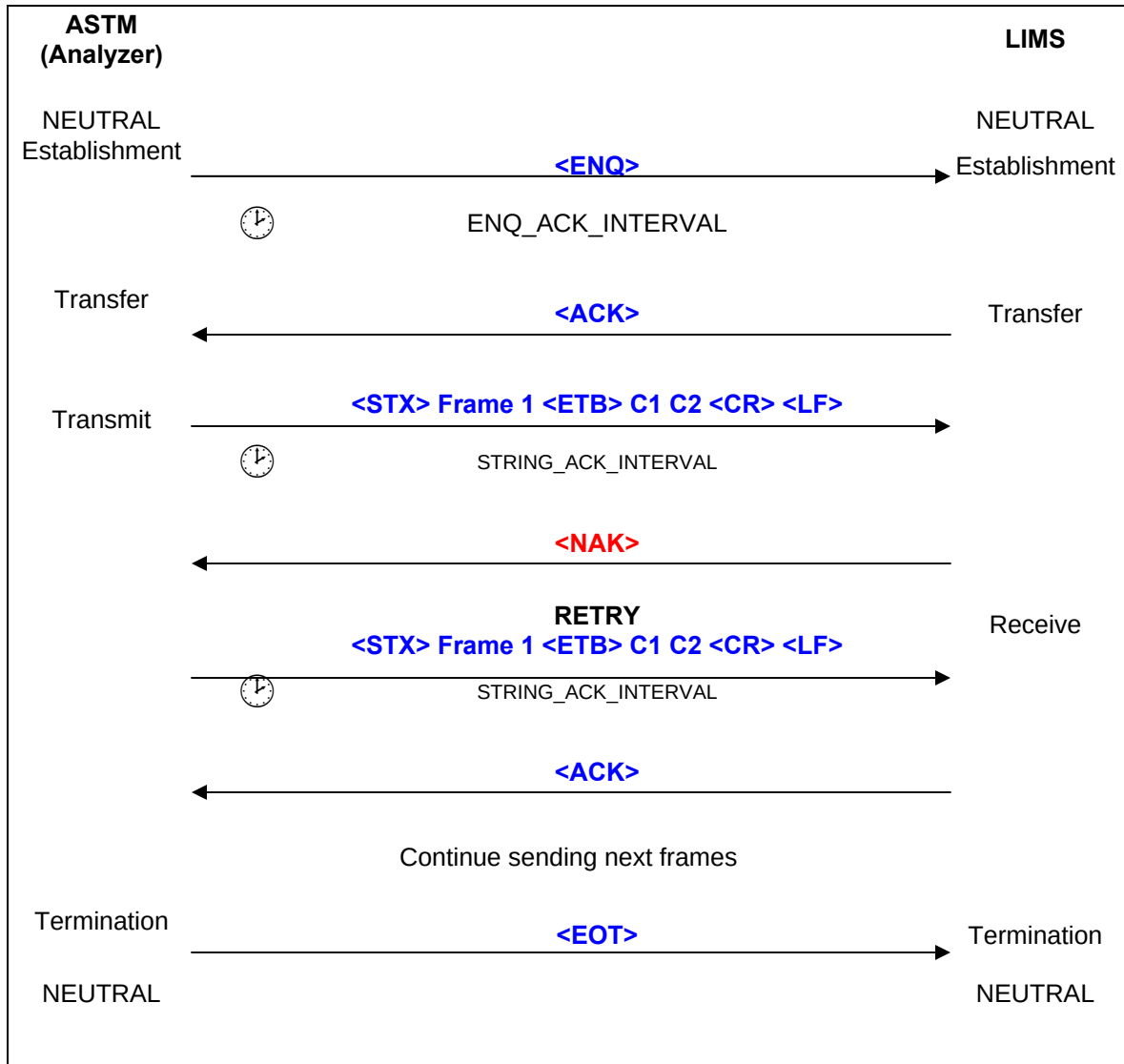
Data Flow Diagram:

Transmitting Message:

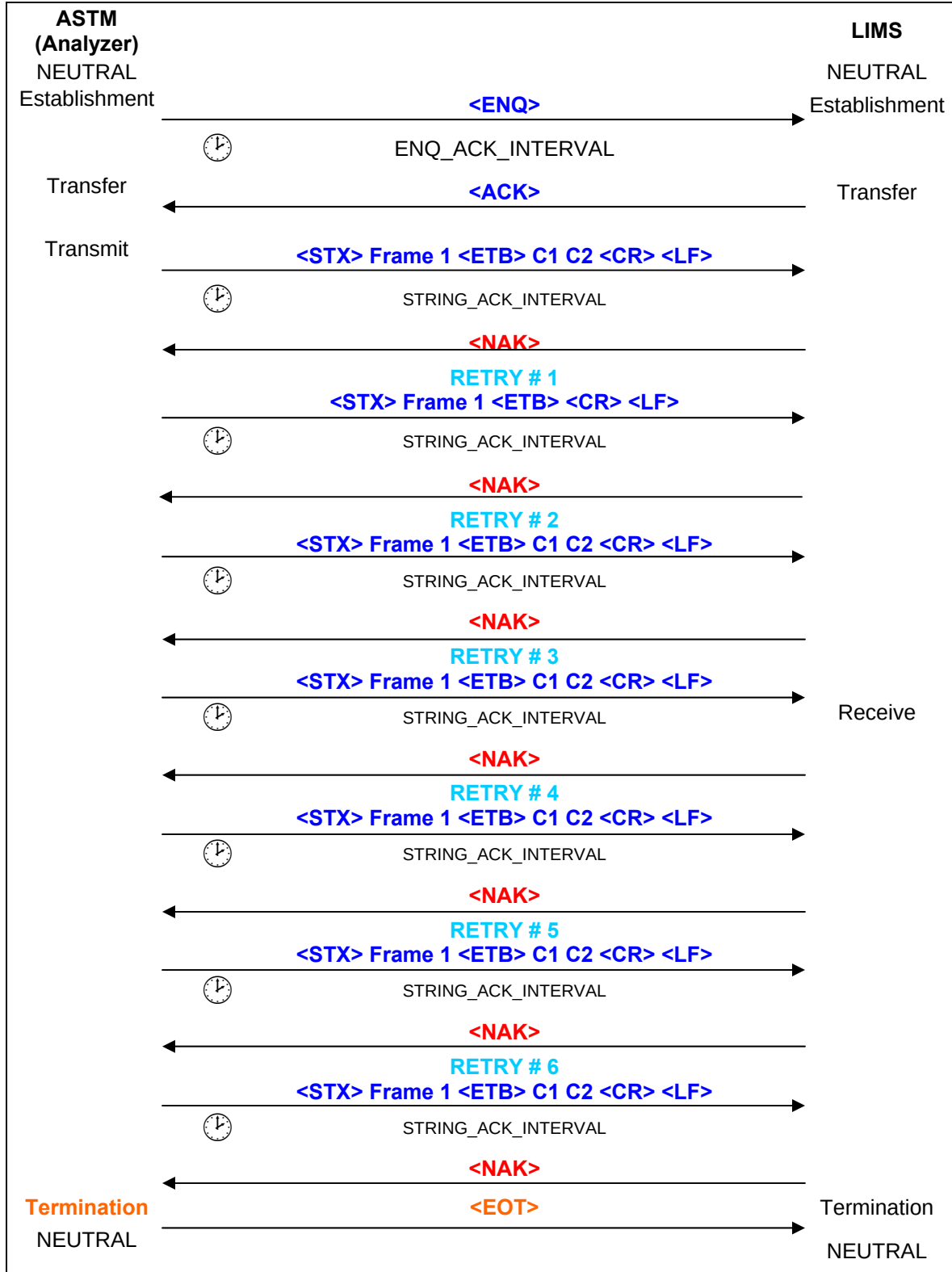
Case 1: Ideal Transmit Data.



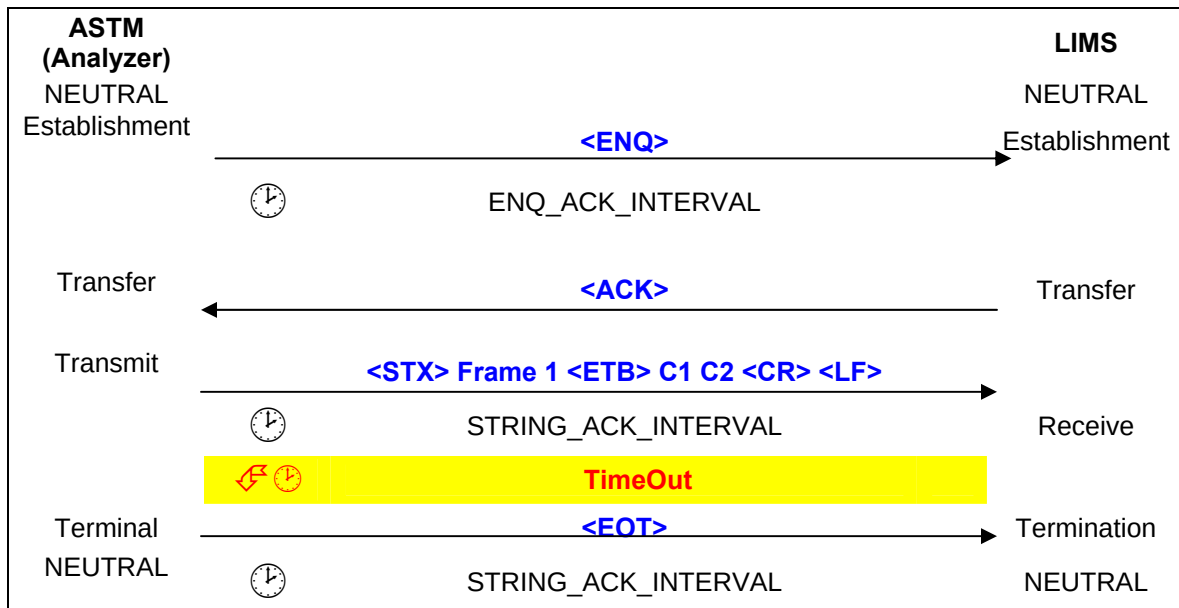
Case 2: <NAK> Received



Case 3: <NAK> Received for more than 6 times.

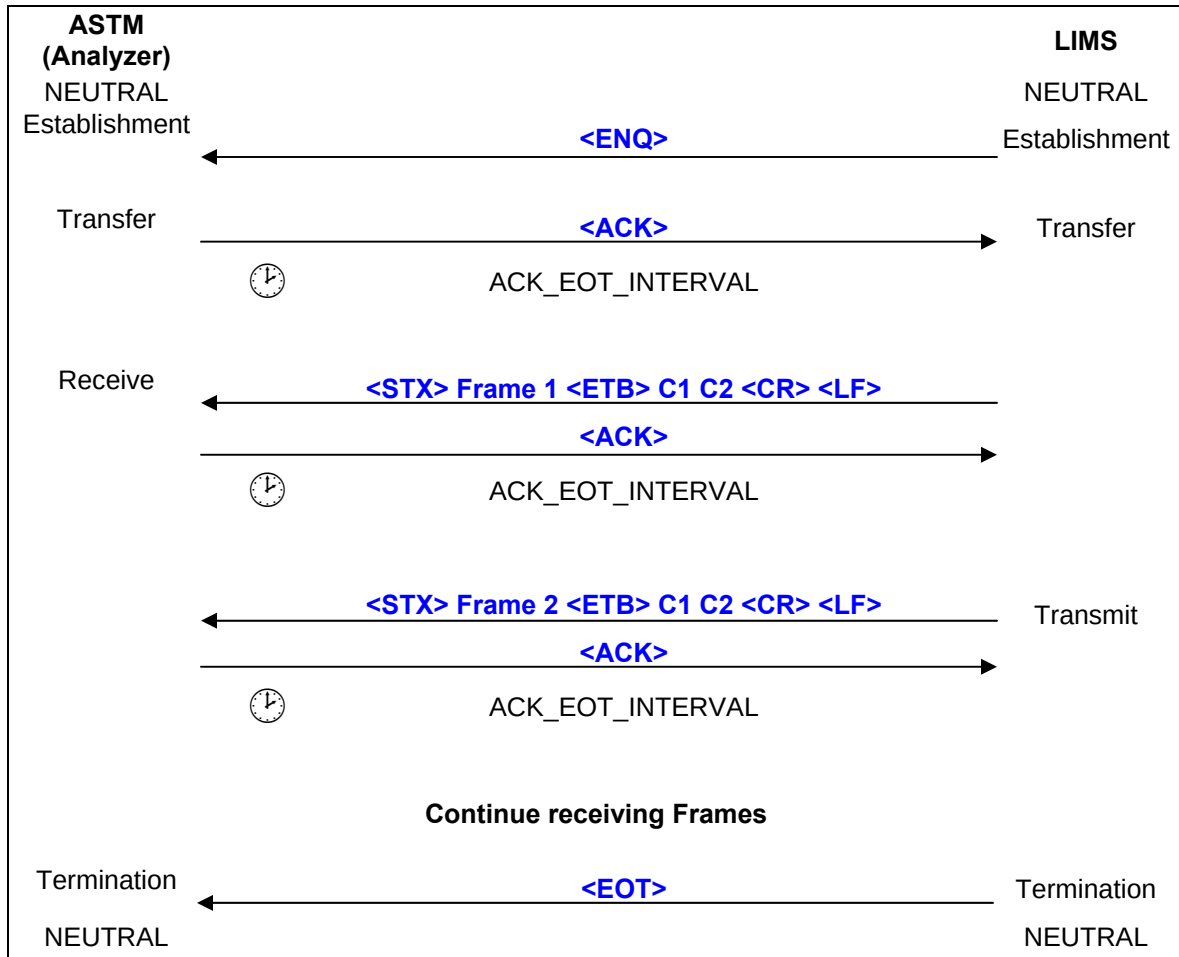


Case 4: Time out after sending frame.

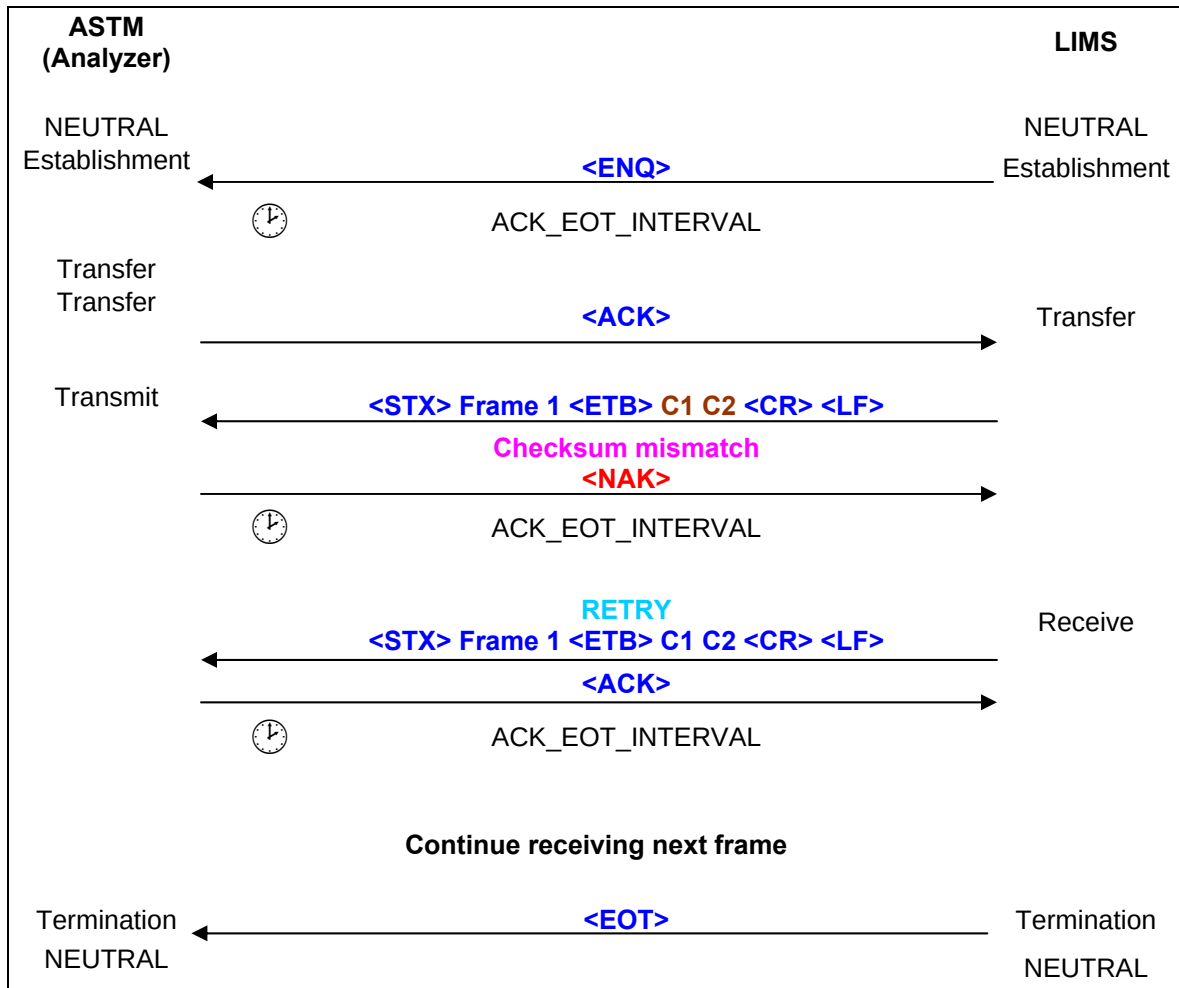


Receiving Message:

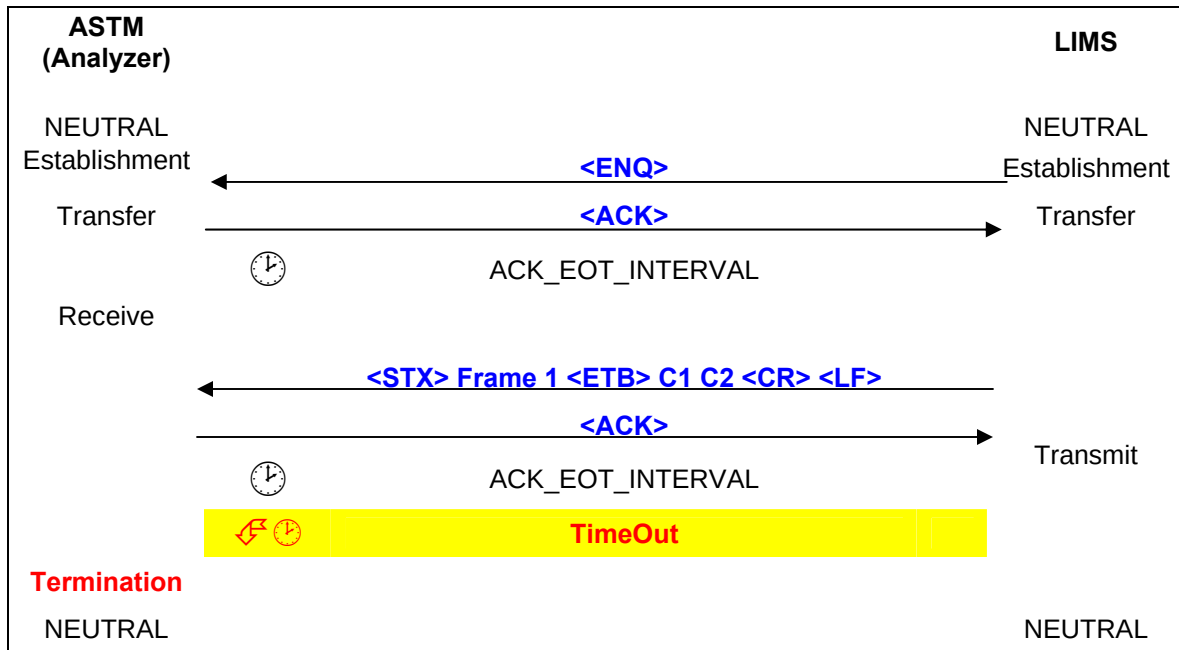
Case 1: Ideal Receive Data



Case 2: Checksum Mismatch



Case 3: Time Out after sending <ACK>



NOTE:

<EOT> and <ENQ> should never get transmitted together.
 There should be some Time delay between <EOT> of previous transmitted DataPacket and <ENQ> of next Data Packet.

CHECKSUM CALCULATION:

Consider a message Frame to be transmitted:

<STX> FN Text Message <ETB> C1 C2 <CR> <LF>

Where:

- C1 & C2 Represents the high nibble (= most significant 4 bit) respectively, the low nibble (=least significant 4 bit) of the 8-bit checksum.
- C1 and C2 are represented as two digits of hex numbers.
- The checksum is the modulus 8 of the sum of ASCII values of the frame characters starting with and including 'FN' and completing with <ETX> respectively <ETB>.

Sample VB Code to calculate Checksum:

"strString" = FN Text Message <ETB>

strString consists of entire message string starting from Frame Number "FN" up till <ETB> or <ETB>.

```
For i = 1 To Len(strString)
    Checksum = Checksum + Asc(Mid$(strString, i, 1))
Next i

strCksm = Right$(Hex$(Checksum), 2)
If Len(strCksm) < 2 Then
    For i = Len(strCksm) To 1
        strCksm = "0" + strCksm
    Next i
End If
```

strCksm now holds the two byte checksum to be transmitted.

SPECIAL FUNCTION CHARACTER TABLE:

Decimal	Hex	Character
0	&H0	<NUL>
1	&H1	<SOH>
2	&H2	<STX>
3	&H 3	<ETX>
4	&H4	<EOT>
5	&H5	<ENQ>
6	&H6	<ACK>
7	&H7	<BEL>
8	&H8	<BS>
9	&H9	<TAB>
10	&HA	<LF>
11	&HB	<VT>
12	&HC	<FF>
13	&HD	<CR>
14	&HE	<SO>
15	&HF	<SI>
16	&H10	
17	&H11	<DC1>
18	&H12	<DC2>
19	&H13	<DC3>
20	&H14	<DC4>
21	&H15	<NAK>
22	&H16	<SYN>
23	&H17	<ETB>
24	&H18	<CAN>
25	&H19	
26	&H1A	<SUB>
27	&H1B	<ESC>
28	&H1C	<FS>
29	&H1D	<GS>
30	&H1E	<RS>
31	&H1F	<US>

ASTM HIGHER LEVEL CONCEPT:

ASTM data is sent or received in terms of packets. Packet starts with the Header (H) and ends with the Terminator (L). Packet without header and terminator is treated as invalid and will be ignored.

ASTM Record Types :

Seq No.	Record Type	Convention	Used
1	Message Header Record	H	Y
2	Patient Information Record	P	Y
3	Test Order Record	O	Y
4	Result Record	R	Y
5	Comment Record	C	Y
6	Request Information Record	Q	Y
7	Scientific Record	S	N
8	Manufacturer Information Record	M	N
9	Message Terminator Record	L	Y

Example :

```
H|^&||Password1|Micro1||||LSI1||P|139-94|19890501074500<CR>
P|1||52483291<CR>
O|1|5762^01|^BC ^POSCOMBO|||||BL|||||F<CR>
R|1|^ORG#|103^Group D Entero<CR>
R|2|^AM^MIC|>16<CR>
R|3|^AM^INTERP1|++<CR>
R|4|^AM^DOSAGE1|PO 250.500 mg Q6h<CR>
.
.
.
R|90|^BIOTYPE|102-34021<CR>
L|1<CR>
```

Each record type has predefined fields. If any field has no data, it should be separated with delimiter. All the delimiters are defined in header.

This message is then passed to ASTM Lower level to be further spitted in to frames of specified lengths and transmitted to LIMS.

ASTM RECORDS:

* Fields mentioned in bold are mandatory.

1	Message Header Record	
	Fields	String format
1	Record Type ID	H
2	Delimiter Definition	`^&
3	Message Control ID	Unique Number
4	Access Password	*****
5	Sender Name Or ID	Manufacturer^Instrument1^Software or firmware version`Instrument2
6	Sender Street Address	Street Address^City^State^Zip^Country Code
7	Reserved Field	
8	Sender Telephone Number	Phone1`Phone2`Phone3 (It may contain areacode ,countrycode,beeper number, hours to call) e.g. +912212345678`+912212345679
9	Characteristics of sender	Contains Parity,Checksums,operational protocols
10	Receiver ID	The name or Id of receiver
11	Comment /Special Instructions	
12	Processing ID	P/T/D/Q (At a time only one , default P)
13	Version No.	ASTM version No. 1394-97
14	Date and Time of message	current datetime YYYYMMDDHHMMSS
15	Carriage Return	<CR> End of the string

H`^&||PSWD|Harper Labs|2937 Southwestern Avenue^Buffalo^NY^73205||319 412-9722||||P|1394-97|19890314<CR>

About Processing ID :

P–Production: Treat message as an active message to be completed according to standard processing.

T–Training: Message is initiated by a trainer and should not have an effect on the system.

D–Debugging: Message is initiated for the purpose of a debugging program.

Q–Quality Control: Message is initiated for the purpose of transmitting quality control/quality assurance or regulatory data.

About Delimiters :

i) Default Delimiters in Header (Point 2)

	Field Delimeter (Alt + 124)
`	Repeat Delimeter (Alt + 96)
^	Component Delimeter (Alt + 94)
&	Escape Delimeter (Alt + 38)
␣	<CR> (Alt + 13)

ii) Delimiters can be changed .

iii) Allowed characters for delimiters. Alt + (7,9,11,12,13,32-126,128-254)

iv) Disallowed characters for delimiters : 0 - 6 ,8,10, 14 - 31 , 127 ,255

v) 13 is reserved for a record terminator (CR) and can not be used for any other purpose.

vi) All delimiters should have different values.

viii) Alphanumeric characters should not be used as delimiters.

2	Patient Record	
	Fields	String format
1	Patient Record Number	P
2	Sequence Number	Frame Number (only 1 digit)
3	Practice Assigned Patient ID	Patient ID
4	Laboratory Assigned Patient Id	
5	PatId3	
6	Patient Name	Name of the Patient (Last Name^First Name^Middle Name^Title) . If Patient Name contains single quote i.e." ' " then it will get replaced by " ` " while saving data.
7	Mother's maiden Name	Required to distinguish between patients with same birthdate and last name. This will be represented only with mother's maiden's surname
8	BirthDate	YYYYMMDDHHMMSS
9	Patient Sex	M/F/U (Male/Femal/Other)
10	Patient Race - Ethnic Origin	W`B`O`NA`H
11	Patient Address	Street Address^City^State^Zip^Country Code If Patient Address contains single quote i.e." ' " then it will get replaced by " ` " while saving data.
12	ReservedField	
13	Patient Telephone No.	Phone1`Phone2`Phone3 (It may contain areacode ,countrycode,beeper number,hours to call) e.g. +912212345678`+912212345679
14	Attending Physician ID	(Ordering Physician`Attending Physician`Referring Physician) Physician IDNumber^Code (401-0^Merchant^Darshak^V`401-0^Mehta^Kantilal^^Shri) If Physician ID contains single quote i.e." ' " then it will get replaced by " ` " while saving data.
15	SpecialField1	
16	SpecialField2	
17	Height	Height/Weight and Unit are separated by component delimiter. 1.2^mtr (default unit is cms for ht and for wt kg).
18	Weight	Unit (mtr/kg) is optional , if the value is in default unit.
19	Patient's known or Suspected Diagnosis	Either ICD -9 code or free text. Multiple diagnoses are separated by repeat delimiter.
20	Patient's Active Medications	Generic name can be used.
21	Patient's Diet	Free text. Indicate conditions that affect results of testing.
22	Practice Field No. 1	
23	Practice Field No. 2	
24	Admission and Discharge dates	Separated by repeat delimiter
25	Admission Status	OP/PA/IP/ER (Only 1 is allowed)
26	Location	Clinic location /Nursing Unit /Ward /Bed If Location contains single quote i.e." ' " then it will get replaced by " ` " while saving data.

27	Nature of Alternative Diagnostic Code and Classifiers	Identifies the class code
28	Alternative Diagnostic Code and Classification	Separated by Repeat delimiter
29	Patient Religion	P/C/M/J/L/H (Only 1)
30	Marital Status	M/S/D/W/A (Only 1)
31	Isolation Status	ARP/BP/ENP/NP/PWP/RI/SE/SI/WSP (Only 1)
32	Language	Patient's primary language
33	HospitalService	Both Code and text seperated by component delimiter
34	HospitalInstitution	
35	DosageCategory	A/P1/P2 (Only 1)
36	Carriage Return	<CR>

P|1|2462|158|287-17-2791|POHL^ALLEN^M|Samules|9600401|M|W|4526 C
street^Fresno^CA^92304 |(402)782-342X242|542^Dr.Brown|||72^in|175^lb||Penicillin|||
19890428 |IP|Ward1|||C|M|WSP||ERR|PC ^ Prompt Care<CR>

About Admission Status :

OP (outpatient)
PA (preadmit)
IP (inpatient)
ER (emergency room).

About Patient Origin :

W white
B black
O asian/pacific islander
NA native american/alaskan native
H Hispanic

About Patient Religion :

P Protestant
C Catholic
M Church of the Latter Day Saints (Mormon)
J Jewish
L Lutheran
H Hindu

About Marital Status :

M married
S single
D divorced
W widowed
A separated

About Isolation Status :

ARP antibiotic resistance precautions
BP Blood and needle precautions
ENP Enteric precautions
NP Precautions for neutropenic patient
PWP Precautions for pregnant women
RI Respiratory isolation
SE Secretion/excretion precautions
SI Strict isolation

WSP Wound and skin precautions

About Dosage Category :

A-ADULT

P1-PEDIATRIC (1-6 months)

P2-PEDIATRIC (6 months-3 years).

3	Test Order Record	
	Fields	String format
1	Test Order Identifier	O
2	Sequence Number	Frame No.
3	Speciman ID	Sample ID^Container No. (Samp1^01)
4	Instrument Speciman ID	Unique identifier assigned by the instrument
5	Universal Test ID	Test Name (^^ALB^^ALP^^LIVER)
6	Priority	S`A`R`C`P S : Stat (Immediate) A : As soon as possible R : Routine C : Callback P : PreOperative
7	Requested/Ordered Date And Time	Denote the date and time , the test order should be considered ordered. Future time is recorded here. (YYYYMMDDHHMMSS)
8	Speciman collection Date and Time	Actual date and time , the sample was collected (YYYYMMDDHHMMSS)
9	Collection End Time	End date and time of a timed speciman collection (YYYYMMDDHHMMSS)
10	Collection Volume	Sample Volume^Unit (default in ml)
11	Collector ID	The person and facility which collected the speciman. If Collector ID contains single quote i.e." ' " then it will get replaced by " ` " while saving data.
12	Action Code	C : Cancel request for the battery or tests named (Delete Test) A : Add the requested tests or batteries to the existing sample N : New requests accompanying a new sample P : Pending sample (Add but don't schedule) L : Reserved (Not in use) X : Sample or test already in process Q : Treat a sample as a Q/C test sample
13	Danger Code	Represents by either test or code , indicating any special hazard associated with the sample
14	Relevant clinical information	Additional information about the sample
15	Date/ Time Speciman Received	Date and Time recorded by laboratory

16	Speciman Descriptor	Sample Type ^ Sample Source Sample Type : Blood , Urine , Serum Sample Source : Sample source body site (left arm ,left hand , right lung)
17	Ordering Physician	Physician IDNumber^Code (401-0 ^ Thomas ^ John^V`401-0^Mehta^Kantilal^^Shri)
18	Physician's Telephone Number	Phone1`Phone2`Phone3 (It may contain area code ,country code, beeper number, hours to call) e.g. +912212345678`+912212345679
19	User Field No.1	
20	User Field No.2	
21	Laboratory Field No.1	
22	Laboratory Field No.2	
23	Date/ Time Result reported or Last modified	
24	Instrument charge to computer system	Accounting reference by the instrument for the tests performed
25	Instrument Section ID	In the case where the multiple instruments are on a single line or a test was moved from one instrument to another , this field will show which instrument or section of an instrument performed the test
26	Report Types	O : Order Record , User asking analysis be performed C : Correction of previously transmitted results (When sending results from Result Reprint) P : Preliminary results (Before Rerun) F : Final Results X : Order Cancelled (When order can not be performed) I : In Instrument Pending Y : No order on Record for this test (in response to query) Z : No record of this patient (In response to query Q : Response to a request-information query
27	Reserved Field	
28	Location or Ward of Speciman Collection	Ward of sample collection
29	Nosocomial Infection Flag	Hospital acquired infection
30	Speciman Service	Specific service responsible for the sample collection
31	Speciman Institution	Institution of sample collection
32	Carriage Return	<CR>

O|1|5762^01||^BC^BloodCulture^POSCOMBO|R|19890501530|198905020700||456^Fam
sworth|N||198905021130|BL^Blood|123^Dr.Wirth|||||Instrument#1||ERR|N<CR>

4	Result Record	
	Fields	String format
1	Result Record Identifier	R
2	Sequence Number	Frame No.
3	Universal Test ID	Test Name (^^ALB) or (^^LIVER^ALB)
4	Data or Measurement Value	Result value
5	Units	ISO 2955
6	Reference Ranges	Lower limit to Upper limit
7	Result Abnormal Flags	L: Below Low Normal H: Above High Normal LL: Below Panic Normal HH: Above Panic High <:Below absolute low >:Above absolute high N: Normal A: AbNormal U: Significant Change Up D: Significant Change Down B: Better, Use when direction not relevant or not defined W: Worse, Use when direction not relevant or not defined
8	Nature of Abnormality Testing	A :Age based population S :Sex based population R :Race based population N :Generic Normal Range was applied to all patients
9	Result Status	C : Correction of previously transmitted results (Patient Report) P : Preliminary results (Before Rerun) F : Final Results X : Order Cancelled (When test can not be preformed) I : In Instrument Pending S : Partial Results M : MIC Level Results R : Result was previously transmitted

		N : This result contains necessary information to run a new order
		Q : Response to a request-information query
		V: Operator verified / approved result
		W: Validity is questionable
10	Date of change in instrument Normative Values or Units	
11	Operator Identification	1st Component : Instrument Operator who performed the test
		2nd Component : Verifier for the test
12	Date / Time Test Started	YYYYMMDDHHMMSS
13	Date / Time Test Completed (Result Date)	YYYYMMDDHHMMSS
14	Instrument Identification	Identifies instrument or section of instrument that performed the test
15	Carriage Return	<CR>

R|1|^^^ALB|2.3|mg/dl<CR>

5	Comment Record	
	Fields	String format
1	Comment Record Identifier	C
2	Sequence Number	Frame No.
3	Comment Source	Comment Origination Point
		P : Practice
		L : Computer System (LIS)
		I : Instrument (ASTM)
4	Comment Text	Code ^Comment Text
5	Comment Type	G : Generic/Free text comment
		T: Test Name comment
		P: Positive Test Comment
		N: Negative Test comment
		I: Instrument Flag Comment
6	Carriage Return	<CR>

C|1|L|Notify IDC if tests positive|G<CR>

6	Request Information Record	
	Fields	String format
1	Request Record Identifier	Q
2	Sequence Number	Frame No.
3	Starting Range ID Number	PatID^SampleID1`SampleID2^Manufacturer defined Battery No. OR ALL . Use Repeat delimiter for multiple samples
4	Ending Range ID Number	PatID^SampleID
5	Universal Test ID	^^ALB`^^ALP / ALL
6	Nature of Request Time Limits	S : Sample Collection Date R : Result Test Date
7	Beginning Request Results Date And Time	This field represent either a beginning date and time for which results are being requested or a single date and time. YYYYMMDDHHMMSS`YYYYMMDDHHMMSS`YYYYMMDDHHMMSS
8	Ending Request Results Date And Time	If not null, specifies the ending or latest date and time for which results are being requested. If beginning date consists of multiple dates , then ending date should be null. (YYYYMMDDHHMMSS)
9	Requesting Physician Name	Physician IDNumber^Code (401-0 ^ Merchant ^ Darshak^V`401-0^Mehta^Kantilal^^Shri)
10	Requesting Physician Telephone Number	Phone1`Phone2`Phone3 (It may contain area code ,country code, beeper number, hours to call) e.g. +912212345678`+912212345679
11	User Field No. 1	
12	User Field No. 2	
13	Request Information Status Codes	C : Correction of Previously transmitted results (Patient Report) P : Preliminary Results (PreRerun results , for tests gone for Rerun) F : Final Result X : Request can not be done , request cancelled I : Results pending in instrument S : Request Partial / UnFinalized results M : Result is a MIC Level R : This Result was previously transmitted A : Abort/cancel last request criteria (allows a new request to follow) N : requesting new or edited results only (Modified from patient report or online/offline results but not transmitted to LIMS) O : Requesting test orders and demographics only D : Requesting demographics only (Patient record)
15	Carriage Return	<CR>

Q|1|032989326||ALL<CR>

7	Message Terminator Record	
	Fields	String format
1	Message Terminator Record Identifier	L
2	Sequence Number	1
3	Termination Code	Nil
		N : Normal Termination
		T : Sender Aborted
		R : Receiver Requested Abort
		E : Unknown System Error
		Q : Error in Last request for information
		I : No Information available from last query
		F : Last request for information processed

L|1|N<CR>

Data transferred from ASTM to LIMS :

1) Result Record :

Only **photometric** and **ISE** tests for **patient sample** and **controls** will be sent to LIMS. Standard and Blank results will not be sent to LIMS.

In following cases results will be sent to LIMS

- Online : When Run is on.
- Offline : From Offline Entry module.
- Result Reprint : From Result Reprint module.
- Patient Report : Only modified results from Patient Report will be sent to LIMS.

2) Request Information Record :

After sample barcode scan, the sampleID will be checked in Application Software. If the sampleID is not available then, the request will be sent to LIMS.

3) Comment Record (Optional) :

This is free text, where user can write any extra information about the record, in his/her own language.

In following cases comment will be sent to LIMS

- Invalid Test : IF test sent from LIMS , is not available in Application Software, comment saying test name XXX not available in instrument, will be sent to LIMS.
- Invalid Result Flag : ASTM has some predefined flags. If Application software flag is not matched with ASTM flag, then the actual flag generated from Application software will be sent to LIMS

4) Patient and Test Order Record :

Normally this information comes from the LIMS. But there is provision in Application software, to make entry for patient and sample. This information will be sent from ASTM to LIMS.

Data transferred from LIMS to ASTM :

- 1) **Patient Record** :
All the patient history.
- 2) **Test Order Record** :
The sample information along with the tests schedules / not scheduled.
- 3) **Request Information Record** :
 - i) Results
 - ii) Patient and Test Order, for which, the entry is made in Application software.
- 4) **Comment Record (Optional)** :
 - i) Patient Information
 - ii) Any other information

Examples of each record type :

Example 1 : **Patient Order Data**

Header

Patient Information Record A
Comment Record for Patient A
Order Information Record A1
Order Information Record A2
Patient Information Record B
Order Information Record B1
Order Information Record B2
Order Information Record B3

Message Terminator

Example 2 : **Result Data**

Header

Patient Information Record A
Comment Record for Patient A
Order Information Record A1
Result Information Record A11
Comment Record For Result A11
Result Information Record A12
Result Information Record A13
Order Information Record A2
Result Information Record A21
Result Information Record A22
Patient Information Record B
Order Information Record B1
Result Information Record B11
Result Information Record B12
Comment Record For Result B12

Order Information Record B2
 Result Information Record B21
 Result Information Record B22
 Order Information Record B3
 Result Information Record B31

Message Terminator

Example 3 : **Request Information Data**
Header
 Request Information Record A
 Message Terminator

While Sending / Receiving the string...

1. Every packet should start with header (H) and end with terminator (L).
2. All delimiters should be defined in Header (H). Delimiters should not be part of field data.
3. Each string in the packet should be max 1024* characters in length.
4. Each string should start with record type (H / P / O / Q / C / L) and terminate with <CR>.
5. The record type ID (H / P / O / Q / C / L) is case sensitive.
6. Each string should contain the Record Sequence Number.
7. There should not be <CR> in between the string.
8. Delimiters should not be included for trailing null fields.
 Example: **P|1|Pat1|||<CR>** should be written as **P|1|Pat1<CR>**
9. Fields should be arranged in the positional order specified.
10. The sending system can use null values to indicate no change. This null value should not overwrite existing data in the receiving system.
11. A field containing only a pair of double quotes (Alt + 34) should be treated as an instruction to the receiver that the existing contents pertaining to that field should be deleted.
12. All dates should be in the format of YYYYMMDDHHMMSS.
13. Measurement units should be expressed with ISO 2955.

*Handled by ASTM Lower Level.

Ref No.	Sr No.	Check in System Settings
6.1.1	1	Allowed characters (Delimiters) : 7,9,11,12,13,32-126,128-254
6.1.1	2	Disallowed characters (Delimiters) : 0 - 6 ,8,10, 14 - 31 , 127 ,255
6.1.1	3	13 is reserved for a record terminator.
6.2	4	All fields are variable in length .But clinical instrument can decide the max length of field.
6.4.1	5	Alphanumeric characters should not be used as delimiters.
	6	Any special characters should not be used in clinical instrument application.
6.4	7	All delimiters should have different values.
	8	System settings should be disabled , when communication is on. (Set Flag when communication is on)
Ref No.	Sr No.	Check while building the string
	1	Every packet should start with header (H) and end with terminator (L) .
6.4.7	2	All delimiters should be defined in Header (H)
	3	Each string in the packet should be max 64000 characters in length.
6.5	4	Each string should start with record type (H / P / O / Q / C / L) and terminate with <CR>
6.5	5	The record type ID is case sensitive

6.6.7	6	Each string should contain the Record Sequence Number.
	7	There should not be <Enter> pressed inbetween the string.
6.4.8	8	Delimiters are not included for trailing null fields.
6.4.9	9	Transmitted records may include more fields than are required by a receiving system. When processing a message , the receiving system may ignore any field it does not require.
6.4.9	10	Fields always be transmitted in the positional order specified.
6.4.10.1	11	The sending system can use null values to indicate no change .This null value should not overwrite existing data in the receiving system.
6.4.10.2		
6.4.10.3	12	A field containing only a pair of double quotes (Alt + 34) should be treated as an instruction to the receiver that the existing contents pertaining to that field should be deleted.
6.6.2	13	All dates should be in the format of YYYYMMDDHHMMSS
6.6.4	14	Measurement units should be expressed with ISO 2955 . (Define 2D Array)

ASTM and LIMS Communication :

Sr #	Instrument Software	ASTM	LIMS
1	Scan the Barcode.	Get the barcode detail from instrument and send to LIMS. ⇒	Receive the sample request from ASTM. ↓
			↓
	With the Patient entry option, view the same data.	Receive test order data from LIMS and send to instrument software.	Send the sample order data to ASTM. ←
2	Run the Sample.	Get the results from instrument application and send to LIMS.	Receive the result from the ASTM.
3	Send the results through Result Reprint option to ASTM		
4	Modify result through Patient Report and send to ASTM.		
5	Make offline entry		

1) Scan the barcode and send request to the LIMS

- i) Schedule the patient.
- ii) Start the Run.
- iii) Click on Sample Added button.
- iv) Send request for new samples to LIMS in batch of 5.

The screenshot displays a laboratory instrument's software interface. The main area features a circular diagram of a sample tray with 30 positions, numbered 1 to 30. A central window displays the text "Please Wait Sample Barcode Scan In Process". To the right of the tray diagram, there are temperature readings: RCT 37.1 °C and RGT 16.0 °C. Below the tray diagram, a legend indicates the status of each position: Scheduled (blue), In Process (yellow), Pending (red), and Completed (green). The interface includes a left sidebar with various function buttons like Patient Entry, Test Parameters, Profiles / Calc, QC/Calibrations, Consumables, Status Monitor, Search, Reports, Master, Utility, Service Check, Maintenance, Settings, and Shut Down. On the right, there are panels for PRE-RUN OPT (Disk Change, Reagent Barcode Scan, Sample Barcode Scan, RGT Selective Level Scan, RGT All Level Scan) and RUN OPTIONS (Calibration, Controls, Photometric, ISE Patient). A large red "SAMPLING PAUSED" message is prominently displayed. At the bottom, there are two tables: one for error messages and another for test results.

S.N.	ERROR MESSAGE	EC
050	Dirty Cuvette At Position 5	DC
049	Dirty Cuvette At Position 4	DC
048	Dirty Cuvette At Position 3	DC
047	Dirty Cuvette At Position 2	DC
046	Dirty Cuvette At Position 1	DC
045	Dirty Cuvette At Position 45	DC
044	Dirty Cuvette At Position 44	DC
043	Dirty Cuvette At Position 43	DC
042	Dirty Cuvette At Position 42	DC
041	Dirty Cuvette At Position 41	DC

SN	POS	TEST	RESULT	UNIT	FLAG
004	7	LDH	321.0	U/L	
003	7	CHOL	38155.80	mg/dl	TEC-H...
002	1	LDH	321.0	U/L	
001	1	CHOL	38700.64	mg/dl	TEC-H...

Message Terminated	Neutral Mode
Received Packet: <pre> H ^& P E 1394-97 20080605114422 Q 1 ^10006122 S 0 L 1 N </pre>	Message Packet to be send:
Communication Data Log: <pre> <ENQ><ACK> <STX>1H ^& P E 1394-97 20080605113909<CR>P 1 0 M 0 0<CR>0 1 1 ^LDH SERUM<CR>R 1 ^LDH 321.0 U/L ^DEFAULT N F 20080605113908<CR>C 1 I Instrument Flag <CR>L 1 N<CR><ETX>A8<CR><LF><ACK> <EOT><ENQ><ACK> <STX>1H ^& P E 1394-97 20080605113945<CR>P 1 0 M 0 0<CR>0 1 3 ^LDH SERUM<CR>R 1 ^LDH 321.0 U/L ^DEFAULT N F 20080605113945<CR>C 1 I Instrument Flag <CR>L 1 N<CR><ETX>AB<CR><LF><ACK> <EOT><ENQ><ACK> <STX>1H ^& P E 1394-97 20080605114422<CR>Q 1 ^10006122 S 0<CR>L 1 N<CR><ETX>5A<CR><LF><ACK> <EOT><ACK> </pre>	

Clear
?

H|`^&|||||P|E 1394-97|20080605114422
Q|1|^10006122||S|||||O
L|1|N

v) In Response to this Query LIMS sends "Patient Information" and "Test Order" data for respective SampleID.

2) Patient and Order data sent from LIMS to ASTM.

i) Patient and Order data entered in LIMS, with new sampleID.

Add Patient...			
* Practice Assigned Pat ID:	10006122	Attending Physician ID:	NEHE*RAM
Lab Assigned Patient ID:	LPAT1	Collector ID:	SP1
Patient ID no. 3:	LPAT13	Referred By:	SP2
Patient Name:	VICHARE*PAT1*V	Patient Height(cm):	1.2*M
Mather's Maiden Name:		Patient Weight(kg):	23
Birthdate:	07-04-71	Patient's Diagnosis:	PDIG1
Patient Sex:	M	Sample Remarks:	PACTMED
Patient Race-Ethnic Origin:	W	Patient Remarks:	DIET
Patient Address:	ANDHERI*MAHARASHTRA	Patient's Diet:	PR1
AGE:	RES1	Active Medications:	PR2
Patient Telephone No.:	8756873	Draw Date:	05-06-2008
Disc No.:	694749387948	Discharge Date:	05-06-2008
		Admission status:	OP
		Location:	ANDHERI
		Nature of Alt Diag. Code:	NAldig
		Alternative Diag. Code:	Altdig
		Patient Religion:	H
		Matitl Status:	M
		Isolation Status:	ARP
		Language:	marathi
		Hospital Service:	HpSer
		Hospital Institution:	HpInst
		Dosage Category:	A
Patient Record			
* Specimen ID:	10006122	Total Sample Volume:	200*ml
Instrument Specimen ID:	IPat1	Sample Volume type:	preeta
* Tests Requested:	ABCD1 ALB ALT AMY AP AST CA CACPC CHOL	* Action Code:	II
Change Tests?		Danger Code:	DngC
* Priority:	R	Relevant Clinical Info:	RCInfo
Request/Order Date Time:	05-06-2008	Received Date / Time :	05-06-2008
* Collection Date Time:	05-06-2008	* Specimen Type:	SERUM
Collection End Time:	05-06-2008	Specimen Source:	
		Ordering Physician:	NEHE*RAM
		Physician's Tel. No.:	233245354
		User Field No. 1:	
		User Field No. 2:	
		Laboratory Field No. 1:	LB1
		Laboratory Field No. 2:	LB2
		Location or Ward:	05-06-2008
		Instrument Charge:	566
		Instrument Section ID:	B1
		* Report Type:	0
		Reserved Field:	RES1
		WARD1	
		Hosocomial Infection flag:	NIF
		Specimen Service:	SPSER
		Specimen Institution:	SPIINST
		Test Order Record	
MARKED * ARE MANDATORY FIELDS			
ADD	DEFAULT	Mandatory	CLEAR
MODIFY	SHOW DETAILS	VIEW	CANCEL
GO BACK			

ii) String generated after clicking on (+) button.

```
<ENQ><ACK>
<STX>1H|^&|||****|TBM-LIMS|Seepz|||E-1394-
97|20080605114926<CR>P|1|10006122|LPAT1|LPAT13|VICHARE^PAT1^V|
|19710704|M|W|ANDHERI^MAHARASHTRA|RES1|8756873`694749387948|^N
ENE^RAM|SP1|SP2|1.2^M|23|PDIG1|PACTMED|DIET|PR1|PR2|20080506`2
0080506|OP|ANDHERI|NAldig|Altdig|H|M|ARP|marathi|HpSer|HpInst
|A<CR>C|1|L|Patient
Information|G<CR>O|1|10006122|IPat1|^^^ALT`^^^AMY`^^^LPS|R|200
80506|20080506|20080506|200^ml|preeta|N|DngC|RCInfo|20080506|
SERUM|^NENE^RAM|233245354|||LB1|LB2|20080506|566|B1|0|RES1|WAR
D1|NIF|SPSER|SPIINST<CR>L|1|N<CR><ETX>6F<CR><LF><ACK>
<EOT>
```

LIMS	
Message Terminated	NEUTRAL MODE
Received Packet:	Message Packet to be send:
	H ^& ^**** TBM-LIMS Seepz E-1394-97 20080605114926 P 1 10006122 LPAT1 LPAT13 VICHARE^PAT1^V 19710704 M W ANDHERI^MAHARASHTRA RES1 8756873^694749387948 ^NENE^RAM SP1 SP2 1.2^M 23 PDIG1 PACTMED DIET PR1 PR2 20080506^ 20080506 OP ANDHERI NAItDig AltDig H M ARP marathi HpSer HpInst A C 1 L Patient Information G O 1 10006122 IPat1 ^***ALT^***AMY^***LPS R 20080506 20080506 20080506 200^ml preeta N DngC RCInfo 20080506 SERUM ^NENE^RAM 233245354 LB1 LB2 20080506 566 B1 0 RES1 WARD1 NIF SPSER SPINST L 1 N
<pre> <ENQ><ACK> <STX>1H ^& ^**** TBM-LIMS Seepz E-1394-97 20080605114926<CR>P 1 10006122 LPAT1 LPAT13 VICHARE^PAT1^V 19710704 M W ANDHERI^MAHARASHTRA RES1 8756873^694749387948 ^NENE^RAM SP1 SP2 1.2^M 23 PDIG1 PACTMED DIET PR1 PR2 20080506^ 20080506 OP ANDHERI NAItDig AltDig H M ARP marathi HpSer HpInst A<CR>C 1 L Patient Information G<CR>O 1 10006122 IPat1 ^***ALT^***AMY^***LPS R 20080506 20080506 20080506 200^ml preeta N DngC RCInfo 20080506 SERUM ^NENE^RAM 233245354 LB1 LB2 20080506 566 B1 0 RES1 WARD1 NIF SPSER SPINST<CR>L 1 N<CR><ETX>6F<CR><LF><ACK> <EOT> </pre>	








iii) String received by the ASTM and stored in the instrument software.

Patient Entry(F3)

Test Parameter(F3)

Profiles / Calc(F4)

QC/Calibration(F5)

Consumables(F6)

Status Monitor(F7)

Search(F8)

Reports(F9)

Master

Utility(F11)

Service Check

Maintenance(F12)

Settings

Shut Down

*** Sample ID** : 10006122 ... ☐ Emergency ☒ Barcoded *** Group** 1 **Position** 9

Sample Type : SERUM **Container Type** : TUBE (10 ml)

Sample Vol Type : Normal **Collection Date** : 06-May-2008

Area : ANDHERI **Reg.Date** : 06-May-2008

Ref. Doctor : ... **Analyst** : preeta

Sample Remark : ...

Patient Name : VICHARE PAT1 V **Category** : Male

Age : 449 **Month(s)** : ... **Patient ID** : ...

Height (m) : 1.2 **Weight (kg)** : 23 **Urine vol (ml/24 hrs)** : 0

Address : ANDHERI MAHARASHTRA **Tel. No.** : 8756873,6947493879

Patient Remark : ...

Clear Schedule

Work List

Mask Test(s)

Copy Test(s)

☒ Scheduled / Pending
 ☐ Masked
 ☐ Not Selected
 ☒ Run Performed
 ☐ Calib Expired

Profiles << PRO >>

Calculated Items << >>

Tests

ALB	AMY	ALP	BID	BIT	CAA	CHO	CKMB	CKN	CLO
CO2	CRE	GGT	GLU	GOT	GPT	HDLC	LDH	LDL	MG
MPR	PHO	PRO	TRIG	UA	UREA	DUMMY	PD2	ALT	LPS

Indication : Click on this button to see the last record

←

→

PRINT

SAVE

CLEAR

EDIT

DELETE

3) Results sent from ASTM to LIMS

i) Online

Patient Entry(F2)

Test Parameters(F3)

Profiles / Calc(F4)

QC/Calibrations(F5)

Consumables(F6)

Status Monitor(F7)

Search(F8)

Reports(F9)

Master

Utility(F11)

Service Check

Maintenance(F12)

Settings

Shut Down

SAMPLE TRAY
REAGENT TRAY
REACTION CURVE
BARCODE SCAN

05-06-2008
11:26:00 AM

G 01

01

RCT 36.9 °C
RGT °C

■ Scheduled

■ In Process

■ Pending

■ Completed

PRE-RUN OPT

☐ Disk Change

☐ Reagent Barcode Scan

☐ Sample Barcode Scan

☐ RGT Selective Level Scan

☐ RGT All Level Scan

Select All

RUN OPTIONS

☐ Calibration

☐ Controls

☒ Photometric

☐ ISE Patient

De-Select All

Add Reagent

Add Sample

SR	POS	TEST	TIME
R1S			
01			
02		PAUSE	
03		PAUSE	
04		PAUSE	
05		PAUSE	
06			
07			
08			
09			
10		PAUSE	
11		PAUSE	
12		PAUSE	
13		PAUSE	
14			
15			
16			
R2			
18	7(1)	CHOL	5:42
19			
20	1(1)	CHOL	5:06
21			
22			
23			
24			
25			
26			
27			
28			
29			
30			
31			
32			
33			
34			
35			
36			

S.N.	ERROR MESSAGE	EC
062	Dirty Cuvette At Position 17	DC
061	Dirty Cuvette At Position 16	DC
060	Dirty Cuvette At Position 15	DC
059	Dirty Cuvette At Position 14	DC
058	Dirty Cuvette At Position 13	DC
057	Dirty Cuvette At Position 12	DC
056	Dirty Cuvette At Position 11	DC
055	Dirty Cuvette At Position 10	DC
054	Dirty Cuvette At Position 9	DC
053	Dirty Cuvette At Position 8	DC

SN	POS	TEST	RESULT	UNIT	FLAG
004	7	LDH	321.0	U/L	
003	7	CHOL	38155.80	mg/dl	TEC-H....
002	1	LDH	321.0	U/L	
001	1	CHOL	38700.64	mg/dl	TEC-H....

Result sent to LIMS

Message Terminated	Neutral Mode
Received Packet: H ^& P E 1394-97 20080605113945 P 1 0 M 0 0 0 1 3 ^^^LDH SERUM R 1 ^^^LDH 321.0 U/L ^DEFAULT N F 20080605113945 C 1 I Instrument Flag L 1 N	Message Packet to be send:
Communication Data Log: <ENQ><ACK> <STX>H ^& P E 1394-97 20080605113909<CR>P 1 0 M 0 0<CR>0 1 1 ^^^LDH SERUM<CR>R 1 ^^^LDH 321.0 U/L ^DEFAULT N F 20080605113908<CR>C 1 I Instrument Flag <CR>L 1 N<CR><ETX>A8<CR><LF><ACK> <EOT><ENQ><ACK> <STX>H ^& P E 1394-97 20080605113945<CR>P 1 0 M 0 0<CR>0 1 3 ^^^LDH SERUM<CR>R 1 ^^^LDH 321.0 U/L ^DEFAULT N F 20080605113945<CR>C 1 I Instrument Flag <CR>L 1 N<CR><ETX>AB<CR><LF><ACK> <EOT>	








H|^&|||||||P|E 1394-97|20080605115331
 P|1|0|||||||0|0
 0|1|1|^^^LDH|||||||SERUM
 R|1|^^^LDH|321|U/L|N|F||||20080605120000
 C|1|I|Instrument Flag
 L|1|N

ii) Offline

Result sent from Result Reprint

- 1) Select the results to be sent to LIMS
- 2) Click on Send To Host

Patient Entry(F2)

Test Parameters(F3)

Profiles / Calc(F4)

QC/Calibrations(F5)

Consumables(F6)

Status Monitor(F7)

Search(F8)

Reports(F9)

Master

Utility(F11)

Service Check

Maintenance(F12)

Settings

Shut Down

Patient Report
Result Reprint
Test Statistics
Calibration Trace
Calibration Monitor
Error Log
Reaction Curve

Report Criteria

☒ Latest Batch
 ☐ Date wise

From : 05-Jun-2008
 To : 05-Jun-2008

Report Type

☒ Patients
 ☐ Controls
 ☐ Standards

No. of Test(s) : 6

Send To Host

Print

☐ Print Lab Details

Sample ID	Patient Name	Test	Result	Unit	Flag	Result Date	Curve #
☒ PAT112		CHOL	258.29	mg/dl		08-Apr-2008 05:26:27 PM	694
☐ PAT113		ALB	283.95	g/dl	TEC-H	08-Apr-2008 05:26:45 PM	695
☒ PAT113		PAMY	NA	U/L	R1*	08-Apr-2008 05:27:03 PM	696
☐ PAT113		CHOL	162.70	mg/dl		08-Apr-2008 05:27:21 PM	697
☐ PAT113		GGT	12.3	U/L	TEC-H	08-Apr-2008 05:27:39 PM	698
☒ PAT113		LDH	321.0	U/L		08-Apr-2008 05:27:57 PM	699

Indication : Select Report Criteria

Results received by LIMS

Message Terminated	Neutral Mode
Received Packet: H ^& P E 1394-97 20080605112009 P 1 0 M 0 0 0 1 PAT113 ^^^LDH SERUM R 1 ^^^LDH 321.0 U/L ^DEFAULT N F 20080408052757 C 1 I Instrument Flag L 1 N	Message Packet to be send:
Communication Data Log: <ENQ><ACK> <STX>1H ^& P E 1394-97 20080605112006<CR>P 1 0 M 0 0<CR>0 1 PAT112 ^^^CHOL SERUM<CR>R 1 ^^^CHOL 258.29 mg/dl ^DEFAULT N F 20080408052627<CR>C 1 I Instrument Flag <CR>L 1 N<CR><ETX>C7<CR><LF><ACK> <EOT><ENQ><ACK> <STX>1H ^& P E 1394-97 20080605112009<CR>P 1 0 M 0 0<CR>0 1 PAT113 ^^^LDH SERUM<CR>R 1 ^^^LDH 321.0 U/L ^DEFAULT N F 20080408052757<CR>C 1 I Instrument Flag <CR>L 1 N<CR><ETX>EC<CR><LF><ACK> <EOT>	








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<ENQ><ACK>
<STX>1H|^&|||||P|E 1394-
97|20080605112006<CR>P|1|0|||||M|||||0|0<CR>0|1|PAT112|^^^CHOL||||
|||||SERUM<CR>R|1|^^^CHOL|258.29|mg/dl|^DEFAULT|N|F|||2008040805262
7<CR>C|1|I|Instrument Flag <CR>L|1|N<CR><ETX>C7<CR><LF><ACK>
<EOT><ENQ><ACK>
<STX>1H|^&|||||P|E 1394-
97|20080605112009<CR>P|1|0|||||M|||||0|0<CR>0|1|PAT113|^^^LDH||||
|||||SERUM<CR>R|1|^^^LDH|321.0|U/L|^DEFAULT|N|F|||20080408052757<CR>
C|1|I|Instrument Flag <CR>L|1|N<CR><ETX>EC<CR><LF><ACK>
<EOT>
  
```

IV) Offline Entry

Make offline entry in the application and click on Save.

	Reagent Position	Backup	Offline Results	ReCalculate
Patient Entry(F2) Test Parameters(F3) Profiles / Calc(F4) QC/Calibrations(F5) Consumables(F6) Status Monitor(F7) Search(F8) Reports(F9) Master Library(F11) Service Check Maintenance(F12) Settings Shut Down	Date : 05-Jun-2008 Laboratory : Instrument : Sample ID : 10006122 Sample Type : SERUM Patient Name : VICHARE PAT1 V Age : 150 Day(s) Category : Male Test : ALT Report Name : ALT(GPT) Unit : mg/dl Normal Lower Limit : 2.00 Normal Upper Limit : 3 Result : 2.6857 Flag : —SELECT— Indication : Please enter proper Result.			
	← → ↶ ↷ PRINT SAVE Click to Save EAR EDIT DELETE			

Same data is received by the LIMS.

Message Terminated	Neutral Mode
Received Packet: H ^& P E 1394-97 20080605115619 P 1 0 0 0 0 1 10006122 ^ALT SERUM R 1 ^ALT 2.6857 mg/dl 2 TO 3 N F 20080605120000 C 1 I Instrument Flag L 1 N	Message Packet to be send: H ^& ^TBM-LIMS Seepz E-1394-97 20080605114926 P 1 10006122 LPAT1 LPAT13 VICHARE^PAT1^V 19710704 M W ANDHERI^MAHARASHTRA RES1 8756873^694749387948 NENE^RAM SP1 SP2 1.2^M 23 PDIG1 PACTMED DIET PR1 PR2 20080506^ 20080506 OP ANDHERI NAHDig AltDig H M ARP marathi Hp Ser HpInst A C 1 L Patient Information G O 1 10006122 IPat ^ALT^AMY^LPS R 20080506 20080506 20080506 200^ml preeta N DngC R C Info 20080506 SERUM NENE^RAM 233245354 LB1 LB2 20080506 566 B1 O RES1 WARD1 NIF SPSER SPINST L 1 N
<pre> <ENQ><ACK> <STX>1H ^& P E 1394-97 20080605115619<CR>P 1 0 0 0<CR>0 1 10006122 ^ALT SERUM<CR>R 1 ^ALT 2.6857 mg/dl 2 TO 3 N F 20080605120000<CR>C 1 I Instrument Flag <CR>L 1 N<CR>ETX>E3<CR>LF><ACK> <EOT> </pre>	








Sample BarcodeScan Testing :

(B- Barcoded ; NB- Non Barcoded)

Before Barcode Scan			After Barcode Scan
SamplePos	SampleID	Tests	
2 (NB)	10077127037056 (sample id of pos 6)	CKN(NP),CALC1(ALB,AMY)	Pos replaced by 6 and tests appended as ALP and ISE
8 (B) SAMP123414	SAMP123414	ALB(NP), AMY(NP) ,ISE	ALP appended to same sampleid
10 (B) SAMPLE1147	SAMP10	CKMB,CKN	SampleID changed to SAMPLE1147 (actual barcode)
11 (B) SAMPLE12345	SAMPLE1157 (sample id of pos 15)	ALP,CKN	Sampleid replaced with SAMPLE12345

MultiXL – ASTM Testing (During Run)

Case 1 : During ‘Run’ the scheduled sample is replaced with the other one.

a) Old sample scheduling is in progress. Results are not yet out.

Test Case	Expected Result	Actual Result	Status
Program ‘SAMP21’ at pos 21. Start the Run. When scheduling starts , click on ‘Sample Added’ Replace sample on pos 21 by the other sample, manually. Wait till the LIS sends the requested sample information.	Position 21 should not be replaced by new sample.	Position 21 remains unchanged.	OK

b) Scheduling over and results are out.

Test Case	Expected Result	Actual Result	Status
Program ‘SAMP21’ at pos 21. Start the Run. Wait till schedule gets over. Once the results are out , click on ‘Sample Added’. Replace sample on pos 21 by the other sample, manually. Wait till the LIS sends the requested sample information.	Position 21 should be replaced by the new sample.	Position 21 is replaced by new sample.	OK

Case 2 : Add new tests for the existing sample , from LIS.

Test Case	Expected Result	Actual Result	Status
Program CODE12812 at position 6, with tests ALB,CRE. Scan the barcode. Send few more tests say ALB,ALP,BIT from LIS for the same sample.	Existing test should not be scheduled again. But new tests should be added to the existing sample and get scheduled.	When testorder data is received from LIS , ALB is ignored and ALP, BIT are added to the sample CODE12812.	OK

Case 3 : During Run, emergency sample sent from LIS.

Test Case	Expected Result	Actual Result	Status
Program sample manually. Start the RUN. Send emergency sample from LIS.	Emergency sample should be given high priority.	Emergency sample gets scheduled as soon as the current sample scheduling is over.	OK

Case 4 : Fill all the positions in the sample tray , with the barcoded samples .Then query to LIS and chk if LIS sends sample information to all queried samples.

Test Case	Expected Result	Actual Result	Status
Fill whole sample tray with the barcodeed samples.	ASTM should query for the scanned barcodes and LIS in response should send the patient and testorder information.	ASTM sent query to the LIS for all the scanned samples. And LIS replied with the patient and testorder data, for all the queried samples.	OK
Programme all the samples in LIS			
Scan the barcode.			

Case 5 : Blank / Standard results should not get transferred to the LIS. Control Results if selected in Host Setting , then only get transferred to the LIS.

Test Case	Expected Result	Actual Result	Status
Programme all blank, std, control and patient in the application.	Blank , Standard results should not get transferred to the LIS.	Blank , Standard results are ignored in all cases.	OK
In Host setting screen, set 'Send Control Results' to false.	Only patient results should be sent to the LIS.	Only patient results are sent to the LIS.	OK
In Host setting screen, set 'Send Control Results' to true.	Only patient and control results should be sent to the LIS.	Patient and control both results are sent to LIS.	OK

Case 6 : ISE results should get transferred to ASTM , only if , in system parameter 'ISE' is on.

Test Case	Expected Result	Actual Result	Status
Programme one sample in application.			OK
Send ISE tests from LIS to ASTM.			
a) In System Parameter, put ISE module 'Off'.	ISE tests should not get programmed in application.	ISE tests are not programmed.	
b) In System Parameter, put ISE module 'On'.	ISE tests should get programmed as other photometric tests.	ISE tests got programmed.	

Case 7 : Cancel the test from LIMS.

Test Case	Expected Result	Actual Result	Status
Programme sample SAMP123450 with tests ALB,ALP and CRE			OK
Send from LIS , same sampleid with test ALB and actioncode as 'C'	ALB test should be removed from sample SAMP123450	Schedule for test 'ALB' is removed for the sample SAMP123450.	

Case 8 : Don't send Results with specific Flags (as per selection in Host Settings) to LIS.

Test Case	Expected Result	Actual Result	Status
In host setting , select the flags , for which we want to send the results to LIS			OK
In test parameters , make the appropriate changes to generate those flags.			
Start the RUN.			
Wait till results are out.	Only the selected flag results should get transferred to LIS	Only the selected flag results got transferred to LIS	

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